The closeness relation on the Cauchy reals is the epsilon-approximation relation

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:RC-sim-characterization' (source lines 1292-1294, Theorem 11.3.32); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 11, section 3, theorem offset 31) make the counted theorem print as Theorem 11.3.32, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: the closeness relation is the epsilon-approximation relation on the Cauchy reals (Theorem 11.3.32).

Allowed standard prerequisites (editorial).

- The two relations. u close-epsilon v is the closeness relation generated along the constructors of the Cauchy reals, and u approx-epsilon v is the explicitly defined epsilon-approximation relation.. The statement identifies them.
- Induction principles of the Cauchy reals. To prove a statement about closeness one may induct on its constructors, and to prove one about elements of RC one may induct on rational points and limits.. They are the two inductions used in the two directions at source lines 1297 and 1301.
- Both relations are mere propositions. Each of the two relations is proposition-valued, so an equivalence reduces to a bidirectional implication.. It is the reduction at source line 1296.

Theorem 11.3.32. *For any* $u, v : \mathbb{R}_{\mathsf{c}}$ *and* $\epsilon : \mathbb{Q}_+$ *we have* $(u \sim_\epsilon v) = (u \approx_\epsilon v)$.

Proof. Since both are mere propositions, it suffices to prove bidirectional implication. For the left-to-right direction, we use $\sim$-induction applied to $C(u, v, \epsilon) :\equiv (u \approx_\epsilon v)$. Thus, it suffices to consider the four constructors of $\sim$. In each case,

u and v are specialized to either rational points or limits, so that the definition of $\approx$ evaluates, and the inductive hypothesis always applies.

For the right-to-left direction, we use $\mathbb{R}_{\mathsf{c}}$-induction to assume that u and v are rational points or limits, allowing $\approx$ to evaluate. But now the definitions of $\approx$, and the inductive hypotheses, supply exactly the data required for the relevant constructors of $\sim$. $\square$